

SHETH L.U.J. COLLEGE OF ARTS AND SIR M.V. COLLEGE OF SCIENCE AND COMMERCE

DATA ANALYSIS WITH SAS / SPSS / R

PRACTICAL NO 15

AIM: Generating basic summaries using str() or summary() (R).

OUTPUT:

```
R 4.5.2 - ~/ABHISHEK DINESH SINGH /
File Edit Code View Plots Session Build Debug Profile Tools Help
Source
Console Background Jobs
> # R Script: Generating Basic Summaries
> # Functions: str() and summary()
> # Dataset: NBA Data.csv
>
> # 1. Import Dataset
> nba_df <- read.csv("NBA Data.csv")
> print("--- Data Loaded ---")
[1] "--- Data Loaded ---"
> print(head(nba_df))
  GAME_ID  TEAM_ID  TEAM_NAME  TEAM_ABBREVIATION  TEAM_CITY  HOME_TEAM  MIN  FGM  FGA  FG_PCT  FG3M
1 21200001 1610612764  Wizards                    WAS      Washington Cavaliers 240:00 32 90 0.356 7
2 21200001 1610612739  Cavaliers                    CLE      Cleveland Cavaliers 240:00 36 79 0.456 7
3 21200002 1610612748  Heat                    MIA      Miami Heat 240:00 43 79 0.544 8
4 21200002 1610612738  Celtics                    BOS      Boston Heat 240:00 39 75 0.520 6
5 21200003 1610612747  Lakers                    LAL      Los Angeles Lakers 240:00 38 77 0.494 3
6 21200003 1610612742  Mavericks                    DAL      Dallas Lakers 240:00 40 85 0.471 5
FG3M FG3_PCT FTM FTA FT_PCT ORB  DREB  REB  AST  STL  BLK  TO  PF  PTS  PLUS_MINUS  EFG_PCT  PIE
1 32 0.250 12 20 0.600 18 21 39 26 11 10 12 19 84 -10 0.32 42.60
2 20 0.350 15 22 0.682 18 36 54 22 7 5 20 21 94 10 0.36 57.40
3 16 0.500 26 32 0.813 5 31 36 25 8 5 8 20 120 13 0.43 55.74
4 13 0.462 23 28 0.821 7 34 41 24 4 2 15 23 107 -13 0.39 44.26
5 13 0.231 12 31 0.387 15 31 46 24 6 5 14 21 91 -8 0.38 45.39
6 15 0.333 14 18 0.778 9 31 40 22 9 5 11 25 99 8 0.40 54.61
COVID_FLAG  RESULT  SEASON  WIN_PCT
1 0 0 12 0
2 0 1 12 100
3 0 1 12 100
4 0 0 12 0
5 0 0 12 0
6 0 1 12 100
>
> # 2. USING str() (Structure)
> print("--- OUTPUT OF str() ---")
[1] "--- OUTPUT OF str() ---"
> str(nba_df)
'data.frame': 31340 obs. of 32 variables:
 $ GAME_ID : int 21200001 21200001 21200002 21200002 21200003 21200003 21200004 21200004 2120
0005 21200005 ...
 $ TEAM_ID : int 1610612764 1610612739 1610612748 1610612738 1610612747 1610612742 1610612761
1610612754 1610612743 1610612755 ...
 $ TEAM_NAME : chr "Wizards" "Cavaliers" "Heat" "Celtics" ...
 $ TEAM_ABBREVIATION: chr "WAS" "CLE" "MIA" "BOS" ...
 $ TEAM_CITY : chr "Washington" "Cleveland" "Miami" "Boston" ...
 $ HOME_TEAM : chr "Cavaliers" "Cavaliers" "Heat" "Heat" ...
 $ MIN : chr "240:00" "240:00" "240:00" "240:00" ...
 $ FGM : int 32 36 43 39 38 40 33 37 33 30 ...
 $ FGA : int 90 79 75 77 85 91 78 88 85 ...
 $ FG_PCT : num 0.356 0.456 0.544 0.52 0.494 0.471 0.363 0.474 0.375 0.353 ...
 $ FG3M : int 7 8 6 3 5 6 5 4 7 ...
 $ FG3_PCT : int 32 20 16 13 13 15 17 11 18 25 ...
 $ FTM : int 12 15 26 23 12 14 16 11 5 17 ...
 $ FTA : int 20 22 32 28 31 18 19 16 11 21 ...
 $ FT_PCT : num 0.6 0.682 0.813 0.821 0.387 0.778 0.842 0.688 0.455 0.81 ...
 $ ORB : int 18 18 5 7 15 9 15 9 16 14 ...
 $ DREB : int 21 36 31 34 31 31 27 37 38 33 ...
 $ REB : int 39 54 36 41 46 40 42 46 54 47 ...
 $ AST : int 26 22 25 24 24 22 18 22 19 18 ...
 $ STL : int 11 7 8 4 6 9 12 3 9 13 ...
 $ BLK : int 10 5 5 2 5 5 8 10 5 11 ...
 $ TO : int 12 20 8 15 14 11 10 18 20 15 ...
 $ PF : int 19 21 20 23 21 25 18 16 22 14 ...
 $ PTS : int 84 94 120 107 91 99 88 90 75 84 ...
 $ PLUS_MINUS : int -10 10 13 -13 -8 -2 -2 -9 9 ...
 $ EFG_PCT : num 0.32 0.36 0.43 0.39 0.38 0.4 0.33 0.37 0.33 0.3 ...
 $ PIE : num 42.6 57.4 55.7 44.3 45.4 ...
 $ COVID_FLAG : int 0 0 0 0 0 0 0 0 ...
 $ RESULT : int 0 1 1 0 0 1 0 1 0 ...
 $ SEASON : int 12 12 12 12 12 12 12 12 ...
 $ WIN_PCT : num 0 100 100 0 0 100 0 100 0 100 ...
Environment History Connections Tutorial
R - Global Environment
meets_df 8482 obs. of 8 variables
NBA_Data 31340 obs. of 32 variables
nba_df 31340 obs. of 32 variables
risk_pivot 300000 obs. of 4 variables
survey 726 obs. of 85 variables
survey_clean 726 obs. of 2 variables
survey_df 726 obs. of 85 variables
wide_df 300000 obs. of 4 variables
youtube_tech_channels 130 obs. of 10 variables
yt_df 130 obs. of 10 variables
Files Plots Packages Help Viewer Presentation
New Folder New File Delete Rename More
Home ABHISHEK DINESH SINGH
Name Size Modified
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Diabetes_Prediction.csv 7.4 MB Dec 8, 2025, 10:50 AM
meets.csv 608.7 KB Dec 8, 2025, 12:27 PM
NBA Data.csv 4.5 MB Dec 8, 2025, 12:27 PM
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PRACTICAL NO 13.R 847 B Dec 8, 2025, 12:32 PM
PRACTICAL NO 14.R 867 B Dec 8, 2025, 12:59 PM
S116 ABHISHEK DINESH SINGH PRACTICAL NO 11.pdf 683.8 KB Dec 8, 2025, 11:56 AM
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survey.csv 571.3 KB Dec 8, 2025, 12:05 PM
youtube_tech_channels_20251120_133753.csv 63.4 KB Dec 8, 2025, 12:41 PM
PRACTICAL NO 15.R 867 B Dec 8, 2025, 1:00 PM
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 $ FGA : int 90 79 75 77 85 91 78 88 85 ...
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 $ FG3_PCT : int 32 20 16 13 13 15 17 11 18 25 ...
 $ FTM : int 12 15 26 23 12 14 16 11 5 17 ...
 $ FTA : int 20 22 32 28 31 18 19 16 11 21 ...
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 $ REB : int 39 54 36 41 46 40 42 46 54 47 ...
 $ AST : int 26 22 25 24 24 22 18 22 19 18 ...
 $ STL : int 11 7 8 4 6 9 12 3 9 13 ...
 $ BLK : int 10 5 5 2 5 5 8 10 5 11 ...
 $ TO : int 12 20 8 15 14 11 10 18 20 15 ...
 $ PF : int 19 21 20 23 21 25 18 16 22 14 ...
 $ PTS : int 84 94 120 107 91 99 88 90 75 84 ...
 $ PLUS_MINUS : int -10 10 13 -13 -8 -2 -2 -9 9 ...
 $ EFG_PCT : num 0.32 0.36 0.43 0.39 0.38 0.4 0.33 0.37 0.33 0.3 ...
 $ PIE : num 42.6 57.4 55.7 44.3 45.4 ...
 $ COVID_FLAG : int 0 0 0 0 0 0 0 0 ...
 $ RESULT : int 0 1 1 0 0 1 0 1 0 ...
 $ SEASON : int 12 12 12 12 12 12 12 12 ...
 $ WIN_PCT : num 0 100 100 0 0 100 0 100 0 100 ...
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S116 ABHISHEK DINESH SINGH PRACTICAL NO 14.pdf 630.4 KB Dec 8, 2025, 12:55 PM
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PRACTICAL NO 15.R 867 B Dec 8, 2025, 1:00 PM
```

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Source

```
R 4.5.2 - ~/ABHISHEK DINESH SINGH />
> # 3. USING summary() (Statistical Summary)
> print("---- OUTPUT OF summary() [Before Factor Conversion] ----")
[1] "---- OUTPUT OF summary() [Before Factor Conversion] ----"
> summary(nba_df)
```

GAME_ID	TEAM_ID	TEAM_NAME	TEAM_ABBREVIATION	TEAM_CITY
Min. :21200001	Min. :1.611e+09	Length:31340	Length:31340	Length:31340
1st Qu.:21500228	1st Qu.:1.611e+09	Class :character	Class :character	Class :character
Median :21800456	Median :1.611e+09	Mode :character	Mode :character	Mode :character
Mean :21797639	Mean :1.611e+09			
3rd Qu.:22201004	3rd Qu.:1.611e+09			
Max. :22401230	Max. :1.611e+09			

HOME_TEAM

MIN	FGM	FGA	FG_PCT
Length:31340	Length:31340	Length:31340	Length:31340
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

FG3M	FG3A	FG3_PCT	FTM	FTA	FT_PCT
Min. : 0.00	Min. : 4.00	Min. :0.0000	Min. : 0.00	Min. : 0.00	Min. :0.0000
1st Qu.: 8.00	1st Qu.:23.00	1st Qu.:0.2960	1st Qu.:13.00	1st Qu.:17.00	1st Qu.:0.7040
Median :10.00	Median :30.00	Median :0.3560	Median :17.00	Median :22.00	Median :0.7780
Mean :10.64	Mean :29.66	Mean :0.3572	Mean :17.35	Mean :22.58	Mean :0.7689
3rd Qu.:13.00	3rd Qu.:36.00	3rd Qu.:0.4170	3rd Qu.:21.00	3rd Qu.:27.00	3rd Qu.:0.8400
Max. :29.00	Max. :70.00	Max. :0.8420	Max. :52.00	Max. :64.00	Max. :1.0000

OREB	DREB	REB	AST	STL	BLK
Min. : 0.00	Min. :12.00	Min. :17.00	Min. : 6.00	Min. : 0.000	Min. : 0.000
1st Qu.: 8.00	1st Qu.:29.00	1st Qu.:39.00	1st Qu.:20.00	1st Qu.: 6.000	1st Qu.: 3.000
Median :10.00	Median :33.00	Median :44.00	Median :24.00	Median : 7.000	Median : 5.000
Mean :10.47	Mean :33.27	Mean :43.74	Mean :23.93	Mean : 7.689	Mean : 4.866
3rd Qu.:13.00	3rd Qu.:37.00	3rd Qu.:48.00	3rd Qu.:27.00	3rd Qu.:10.000	3rd Qu.: 6.000
Max. :38.00	Max. :60.00	Max. :81.00	Max. :50.00	Max. :22.000	Max. :20.000

TO	PF	PTS	PLUS_MINUS	EFG_PCT	PIE
Min. : 1.00	Min. : 4.0	Min. : 58.0	Min. : -73	Min. :0.1900	Min. : 2.90
1st Qu.:11.00	1st Qu.:17.0	1st Qu.:98.0	1st Qu.: -9	1st Qu.:0.3700	1st Qu.:43.84
Median :13.00	Median :20.0	Median :108.0	Median : 0	Median :0.4100	Median :50.00
Mean :13.55	Mean :19.9	Mean :107.8	Mean : 0	Mean :0.4159	Mean :49.99
3rd Qu.:16.00	3rd Qu.:23.0	3rd Qu.:117.0	3rd Qu.: 9	3rd Qu.:0.4500	3rd Qu.:56.16
Max. :31.00	Max. :42.0	Max. :176.0	Max. : 73	Max. :0.7800	Max. :97.10

COVID_FLAG	RESULT	SEASON	WIN_PCT
Min. :0.0000	Min. :0.0	Min. :12.00	Min. : 0.00
1st Qu.:0.0000	1st Qu.:0.0	1st Qu.:15.00	1st Qu.:38.16
Median :0.0000	Median :0.5	Median :18.00	Median :50.00
Mean :0.1365	Mean :0.5	Mean :17.97	Mean :50.08
3rd Qu.:0.0000	3rd Qu.:1.0	3rd Qu.:21.00	3rd Qu.:61.90
Max. :1.0000	Max. :1.0	Max. :24.00	Max. :100.00

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```
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> # 4. IMPROVING summary() WITH FACTORS
> # Convert Team column to factor (Categorical)
> nba_df$Team <- as.factor(nba_df$Team)
```

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